Televisions

A television set combines a receiver, display, and speakers to re-create video and audio transmitted by a broadcaster (see pp.188–189). Technological advances have led to slimmer televisions that produce higher-definition pictures and can connect to the Internet.

Flat screens

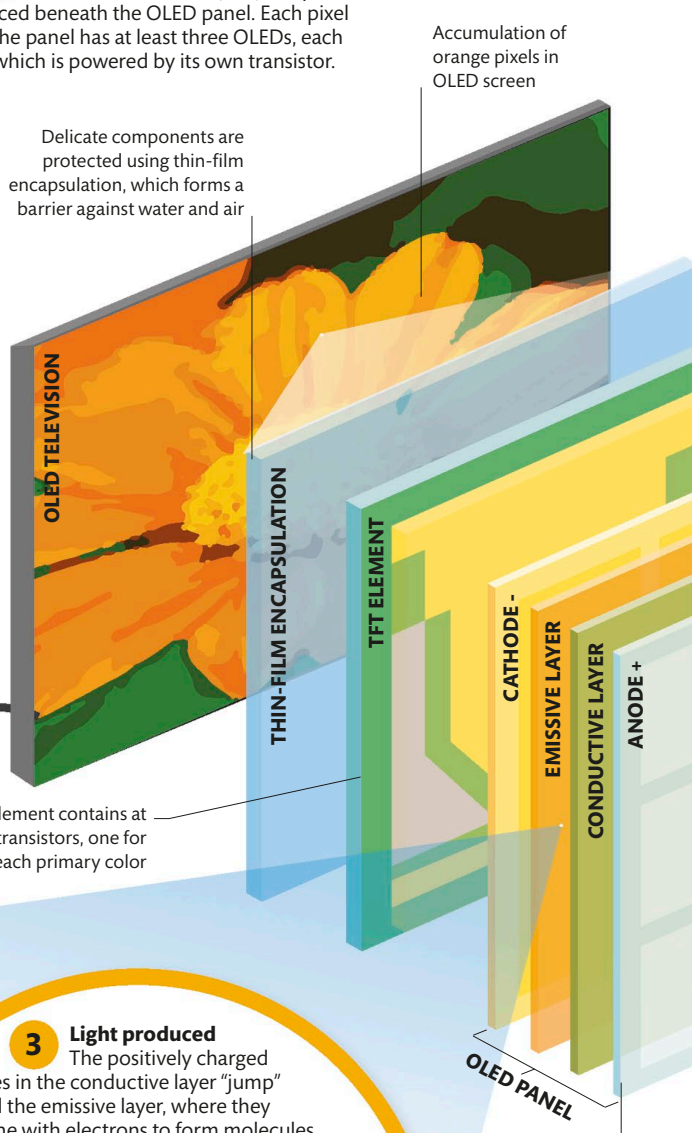
For decades, the only type of television was the cathode-ray tube (CRT) television, which created images using a vacuum tube to deflect beams of electrons onto a screen. These bulky devices have now been replaced by flat-screen televisions. Liquid crystal display (LCD) technology, which uses the optical properties of liquid crystal to create images, is incorporated into flat-screen televisions. In organic light-emitting diode (OLED) flat screens, a layer of organic matter produces light in response to an electric current. Each LED lights up individually so, unlike LCD screens, they do not need a backlight to supply light.

How OLED flat screens work

LEDs emit light when electrons move between a material with more electrons and a material with fewer electrons. OLEDs work similarly but are built using layers of organic (carbon-based) material.

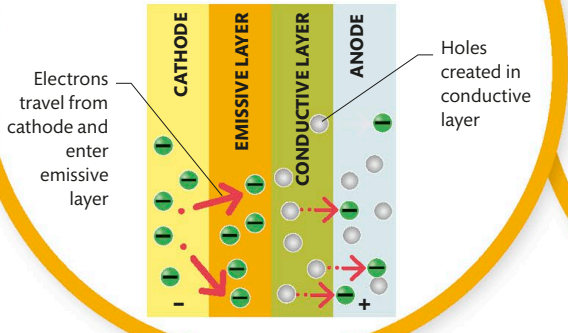
1 Supplying electric charge

A thin-film transistor (TFT) array is placed beneath the OLED panel. Each pixel in the panel has at least three OLEDs, each of which is powered by its own transistor.



2 Migrating electrons

The power source supplies electrons to the cathode and emissive layer, making the latter negatively charged. The anode and conductive layer lose electrons, which leaves behind "holes" and makes the conductive layer positively charged.



3 Light produced

The positively charged holes in the conductive layer "jump" toward the emissive layer, where they recombine with electrons to form molecules. These molecules enter an "excited state," and when they relax, energy is released as light.

